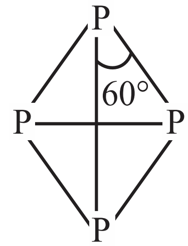
CHAPTER 10

The p-Block Elements

Sr. No.	Concept	Description	Other information
1.	**General electronic configuration of p-Block elements**	ns^2np^{1-6}	
2.	**Group 15 elements:** (a) General electronic configuration (b) Metallic radii (c) General oxidation state	 ns^2np^3 Increases down the group Common oxidation state = + 3	 Other O.S. = +5
3.	**Physical properties of group 15th elements (Nitrogen family)** (a) Covalent radii (b) Ionic radii (c) Electronegativity (d) Ionization enthalpy (e) Melting point (f) Boiling point (g) Density	 $N < P < As < Sb < Bi$ $Sb < Bi < N < P < As$ $Sb \approx Bi < As < P < N$ $Bi < Sb < As < P < N$ $N < P < Bi < Sb < As$ $N < P < As < Bi < Sb$ $N < P < As < Sb < Bi$	 Covalent radii increases down the group Density increases down the group

Sr. No.	Concept	Description	Other information
4.	**Chemical properties of Group 15 elements**		
	(a) Halides	General formula, EX_3 or EX_5.	
	(i) Stability	$NF_3 > NCl_3 > NBr_3$	Due to large difference in size.
	(ii) Lewis acid strength	$PCl_3 > AsCl_3 > SbCl_3$ & $PF_3 > PCl_3 > PBr_3 > PI_3$	Decreasing electron density on N as the electronegativity of halogen increase.
	(iii) Lewis base strength	$NI_3 > NBr_3 > NCl_3 > NF_3$	
	(iv) Bond angle	$PF_3 < PCl_3 < PBr_3 < PI_3$	Decreasing b.p. - b.p. repulsions.
	(b) Hydrides		
	(i) Bond angle	$NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$	
	(ii) Basic character	$NH_3 > PH_3$ (AsH_3, SbH_3 and BiH_3 and not basic)	
	(iii) Boiling point	$PH_3 < AsH_3 < NH_3 < SbH_3 < BiH_3$	
	(iv) Melting point	$PH_3 < AsH_3 < SbH_3 < NH_3$	
	(v) Thermal stability	$NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$	
	(vi) Reducing character	$NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$	
	(vii) Solubility	$\underbrace{NH_3,}_{\text{(Soluble)}}\ \underbrace{PH_3, AsH_3, SbH_3, BiH_3}_{\text{(Insoluble)}}$	
	(c) Oxides	General formula , E_2O_3 and E_2O_5.	
	(i) Oxides of Nitrogen		
	• Acidic strength of trioxides	$N_2O_3 > P_4O_6 > As_4O_6 > Sb_4O_6$	Electronegativity of central atom decreases
	• Acidic strength of oxides of nitrogen	$\underbrace{N_2O < NO}_{\text{Neutral}} < N_2O_3 < N_2O_4 < N_2O_5$	O. No. increases

<table>
<tr><th>Sr. No.</th><th>Concept</th><th>Description</th><th>Other information</th></tr>
<tr><td></td><td>(d) Allotropic forms of Phosphorus
(i) White Phosphorus</td><td>



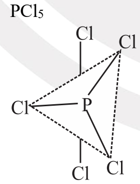
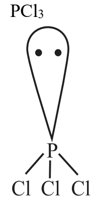
- Translucent white waxy solid.
- Poisonous & show chemiluminescence.
- Soluble in CS_2 but insoluble in H_2O.
- Occurs in descrete units.
- Highly reactive due to angle strain.
- Fumes in air due to formation of P_4O_{10}.

</td><td></td></tr>
<tr><td></td><td>(ii) Red Phosphorus</td><td>

- Grey luster.
- Insoluble in water and CS_2.
- Nonpoisonous.
- No chemiluminescence.
- Obtained by heating white P at 573 K.
- Occurs as polymer.
- So it is less reactive.

</td><td></td></tr>
<tr><td></td><td>(iii) Black Phosphorus</td><td>

- α-black phosphorous - Prepared by heating Red P at 803 K
- β-black phosphorous - Prepared by heating white P at 473 K

</td><td></td></tr>
<tr><td></td><td>(iv) Halides of Phosphorus</td><td>
<table>
<tr><th>Compound</th><th>Preparation</th><th>Properties</th></tr>
<tr><td>PCl_5
</td><td>$P_4 + 10Cl_2 \rightarrow 4PCl_5$
(white or red)
$P_4 + 10SO_2Cl_2 \rightarrow 4PCl_5 + 10SO_2$</td><td>$PCl_5 + 4H_2O(\text{excess}) \rightarrow H_3PO_4 + 5HCl$
$PCl_5 + SO_2 \rightarrow SOCl_2 + POCl_3$
$6PCl_5 + P_4O_{10} \rightarrow 10POCl_3$
$PCl_5 + Zn \rightarrow ZnCl_2 + PCl_3$</td></tr>
<tr><td>PCl_3
</td><td>$P_4 + 6Cl_2 \rightarrow 4PCl_3$
$P_4 + 8SO_2Cl_2 \rightarrow 4PCl_3 + 4SO_2 + 2S_2Cl_2$</td><td>$PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$
$3CH_3COOH + PCl_3 \rightarrow 3CH_3COCl + H_3PO_3$
$3C_2H_5OH + PCl_3 \rightarrow 3C_2H_5Cl + H_3PO_3$</td></tr>
</table>
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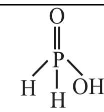
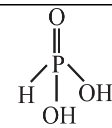
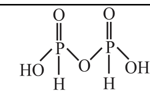
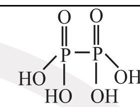
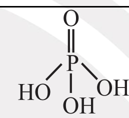
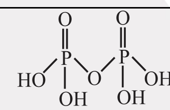
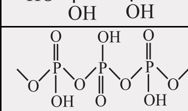
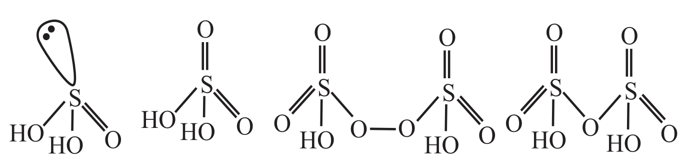
<table>
<tr><th>Sr. No.</th><th>Concept</th><th>Description</th><th>Other information</th></tr>
<tr><td></td><td>(vi) Oxoacids of Phosphorus</td><td>
<table>
<tr><th>Name</th><th>Formula</th><th>Oxidation state of phosphorus</th><th>Structure</th><th>Preparation</th></tr>
<tr><td>Hypophosphorous acid (Phosphinic acid)</td><td>H_3PO_2</td><td>+1</td><td></td><td>white P_4 + alkali</td></tr>
<tr><td>Orthophosphorous acid (Phosphonic acid)</td><td>H_3PO_3</td><td>+3</td><td></td><td>$P_2O_3 + H_2O$</td></tr>
<tr><td>Pyrophosphorous acid</td><td>$H_4P_2O_5$</td><td>+3</td><td></td><td>PCl_3 + H_3PO_3</td></tr>
<tr><td>Hypophosphoric acid</td><td>$H_4P_2O_6$</td><td>+4</td><td></td><td>Red P_4 + alkali</td></tr>
<tr><td>Orthophosphoric acid</td><td>H_3PO_4</td><td>+5</td><td></td><td>$P_4O_{10} + H_2O$</td></tr>
<tr><td>Pyrophosphoric acid</td><td>$H_4P_2O_7$</td><td>+5</td><td></td><td>Phosphoric acid + heat</td></tr>
<tr><td>Metaphosphoric acid</td><td>$(HPO_3)_n$</td><td>+5</td><td></td><td>Phosphorus acid + Br_2, heat in a sealed tube</td></tr>
</table>
</td><td></td></tr>
<tr><td>5.</td><td>Group 16 elements:
(a) General electronic configuration</td><td>ns^2np^4</td><td></td></tr>
<tr><td></td><td>(b) Oxidation state</td><td>Common oxidation state = +2</td><td>Other O.S. = -2, +4, +6</td></tr>
<tr><td></td><td>(c) Stability of O.S.</td><td>(i) Stability of +2 O.S. O > S > Se > Te > Po</td><td>Decreases down the group</td></tr>
<tr><td></td><td></td><td>(ii) Stability of +4 O.S. O < S < Se < Te < Po</td><td>Increases down the group</td></tr>
<tr><td>6.</td><td>Physical properties of group 16th elements (Oxygen family)
(a) Covalent radii</td><td>O < S < Se < Te < Po</td><td>Increases down the group.</td></tr>
<tr><td></td><td>(b) Ionic radii</td><td>O < S < Se < Te < Po</td><td>Increases down the group.</td></tr>
<tr><td></td><td>(c) Ionization enthalpy</td><td>O > S > Se > Te > Po</td><td>Decreases down the group.</td></tr>
</table>

<table>
<tr><th>Sr. No.</th><th>Concept</th><th>Description</th><th>Other information</th></tr>
<tr><td></td><td>(d) Electron affinity
(e) Electronegativity
(f) Melting point
(g) Boiling point
(h) Density</td><td>S > Se > Te > Po > O
O > S > Se > Te > Po
O < S < Se < Po < Te
O < S < Se < Po < Te
O < S < Se < Te</td><td></td></tr>
<tr><td>7.</td><td>Chemical properties of Group 16 elements
(a) Halides
(i) Stability of halides
(b) Hydrides
(i) Boiling point
(ii) Volatility
(iii) Bond angle
(iv) Acidic character
(v) Reducing character
(vi) Thermal stability
(c) Oxides
(i)Simple oxides
(d) Allotropic forms of Sulphur
(i) Rhombic Sulphur [α]
(ii) Monoclinic Sulphur [β]
(e) Oxoacids of Sulphur</td><td>Group 16 form halides of the type EX_6, EX_4 and EX_2.
$F^- > Cl^- > Br^- > I^-$
$H_2S < H_2Se < H_2Te < H_2O$
$H_2S > H_2Se > H_2Te > H_2O$
$H_2O > H_2S > H_2Se > H_2Te$
$H_2O < H_2S < H_2Se < H_2Te$
$H_2Te > H_2Se > H_2S > H_2O$
$H_2O > H_2S > H_2Se > H_2Te$
General formula , EO_2 and EO_3.
<table>
<tr><td>Simple oxides</td><td>MgO, Al_2O_3</td></tr>
<tr><td>Mixed oxides</td><td>Pb_3O_4, Fe_3O_4</td></tr>
<tr><td>Acidic oxides</td><td>SO_2, Cl_2O_7, CO_2, N_2O_5</td></tr>
<tr><td>Basic oxides</td><td>Na_2O, CaO, BaO</td></tr>
<tr><td>Amphoteric oxides</td><td>Al_2O_3</td></tr>
<tr><td>Neutral oxides</td><td>CO, NO, N_2O</td></tr>
</table>
• Exist in room temperature.
• Soluble in CS_2, but insoluble in H_2O.
• Yellow in colour.
• Exist below 369 K.

• Soluble in CS_2.
• Obtained by melting rhombic sulphur above 369 K.

H_2SO_3 Sulphurous acid; H_2SO_4 Sulphuric acid; $H_2S_2O_8$ Peroxodisulphuric acid; $H_2S_2O_7$ Pyrosulphuric acid (Oleum)</td><td></td></tr>
</table>

Sr. No.	Concept	Description	Other information
8.	**Group 17 elements:** (a) General electronic configuration (b) General oxidation state	ns^2np^5 Common oxidation state = -1	Other O.S.= +1, +3, +5, +7
9.	**Physical properties of group 17th elements (Halogen family)** (a) Covalent radii (b) Ionic radii (c) Electronegativity (d) Ionization enthalpy (e) Electron affinity (f) Melting point (g) Boiling point (h) Density	 F < Cl < Br < I F < Cl < Br < I F > Cl > Br > I F > Cl > Br > I Cl > F > Br > I $F_2 < Cl_2 < Br_2 < I_2$ $F_2 < Cl_2 < Br_2 < I_2$ $F_2 < Cl_2 < Br_2 < I_2$	
10	**Chemical properties of Group 17 elements** (a) Halides • The order of reactivity (i) Hydrogen halides • Boiling points • Melting points • Bond lengths • Bond dissociation enthalpy • Acidic strength • Thermal stability • Reducing power (ii) Metal halides • Ionic character (b) Oxides	 $F_2 > Cl_2 > Br_2 > I_2$ HF > HI > HBr > HCl HI > HF > HBr > HCl HI > HBr > HCl > HF HF > HCl > HBr > HI HI > HBr > HCl > HF HF > HCl > HBr > HI HI > HBr > HCl > HF MF > MCl > MBr > MI Fluorine forms two oxides OF_2 and O_2F_2 called oxygen fluorides.	

<table>
<tr><th>Sr. No.</th><th>Concept</th><th>Description</th><th>Other information</th></tr>
<tr><td></td><td>(c) Oxoacids of Halogens</td><td>
<table>
<tr><th colspan="5">Variation of the general properties of oxoacids of halogens</th></tr>
<tr><th>Halogen</th><th>Hypohalous acids (O.S. of halogen = +1)</th><th>Halous acids (O.S. of halogen = +3)</th><th>Halic acids (O.S. of halogen = +5)</th><th>Perhalic acids (O.S. of halogen = +7)</th></tr>
<tr><td>F</td><td>HOF</td><td>–</td><td>–</td><td>–</td></tr>
<tr><td>Cl</td><td>HClO</td><td>$HClO_2$</td><td>$HClO_3$</td><td>$HClO_4$</td></tr>
<tr><td>Br</td><td>HBrO</td><td>–</td><td>$HBrO_3$</td><td>$HBrO_4$</td></tr>
<tr><td>I</td><td>HIO</td><td>–</td><td>HIO_3</td><td>HIO_4</td></tr>
</table>
</td><td></td></tr>
<tr><td></td><td>(d) Interhalogen compounds</td><td>
<table>
<tr><th>Type</th><th>Hybridisation</th><th>Shape</th></tr>
<tr><td>XX</td><td>sp^3</td><td>Linear</td></tr>
<tr><td>XX_3</td><td>sp^3d</td><td>T-shaped</td></tr>
<tr><td>XX_5</td><td>sp^3d^2</td><td>Square pyramidal</td></tr>
<tr><td>XX_7</td><td>sp^3d^3</td><td>Pentagonal bipyramidal</td></tr>
</table>
</td><td></td></tr>
<tr><td>11.</td><td>Group 18 elements:
(a) General electronic configuration</td><td>ns^2np^6</td><td></td></tr>
<tr><td>12.</td><td>Physical properties of group 18th elements (Noble gas family)
(a) Atomic radii
(b) Electron gain enthalpy
(c) Ionization enthalpy
(d) Melting point
(e) Boiling point
(f) Density</td><td>He < Ne < Ar < Kr < Xe
He < Rn < Xe < Ar ≈ Kr < Ne
He > Ne > Ar > Kr > Xe > Rn
Ne < Ar < Kr < Xe <Rn
He < Ne < Ar < Kr < Xe <Rn
He < Ne < Ar < Kr < Xe <Rn</td><td></td></tr>
<tr><td>13.</td><td>Chemical properties of Group 18 elements
(a) Xenon-fluorine compounds

(b) Xenon-oxygen compounds</td><td>$\underset{\text{(excess)}}{Xe} + F_2 \xrightarrow{673K,\ 1bar} XeF_2$; $\underset{\text{(1:5 ratio)}}{Xe} + 2F_2 \xrightarrow{873K,\ 7bar} XeF_4$
$\underset{\text{(1:20 ratio)}}{Xe} + 3F_2 \xrightarrow{573K,\ 60\text{-}70bar} XeF_6$

$6XeF_4 + 12H_2O \longrightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$;
$XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$;
$\left.\begin{array}{l} XeF_6 + H_2O \longrightarrow XeOF_3 + 2HF \\ XeF_6 + 2H_2O \longrightarrow XeO_2F_2 + 4HF \end{array}\right\}$ Partial hydrolysis of XeF_6</td><td></td></tr>
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